

Muhammad Ahmad 2528065 MS(AI). 641 - AI for Robotics

1. Introduction

In this lab, I worked with the **Turtlesim** package in **ROS 2** to simulate and control turtle movements. The objective was to implement various movement behaviors for the turtles, including:

- Moving in a straight line.
- Moving towards a goal.
- Moving in a square and a circle pattern.

Additionally, I created a **ROS 2 launch file** for **Task 1** that publishes and subscribes to complex numbers, allowing the user to set the complex number at launch-time. The lab also required using **rosbag2** for recording topics for later analysis.

Task 1: Launch File for Complex Number Publisher and Subscriber

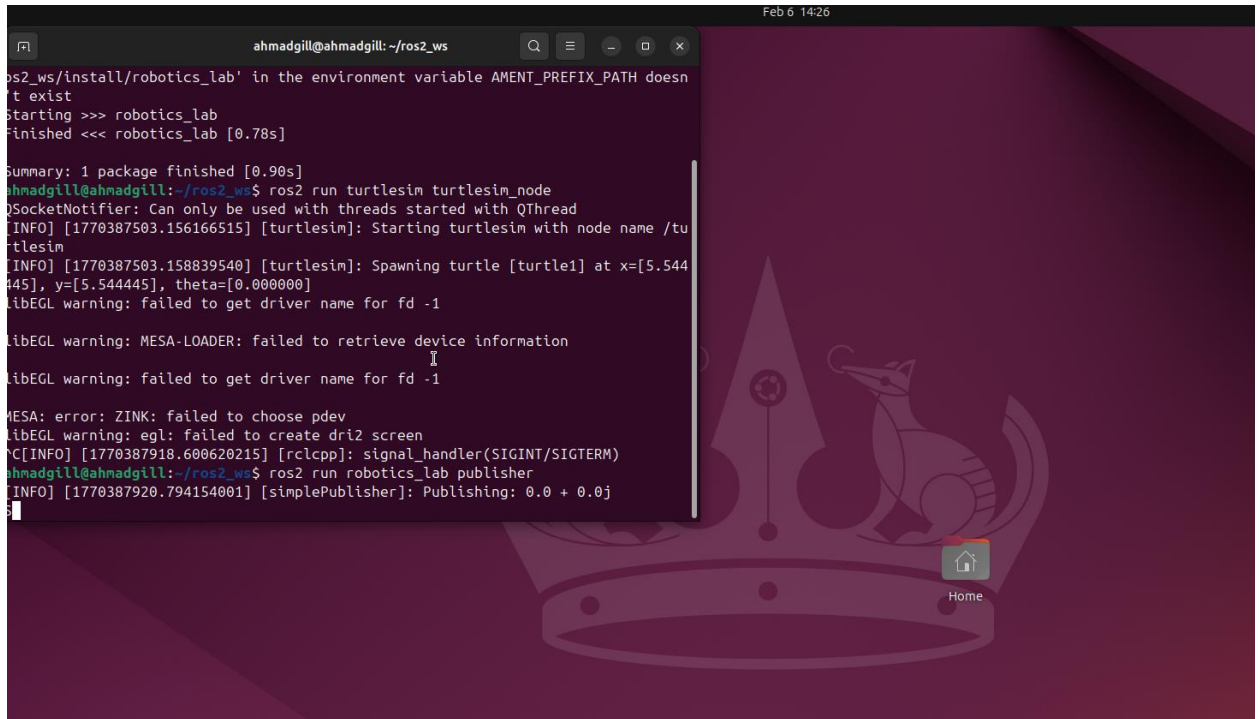
Objective

The objective of Task 1 was to create a **ROS 2 launch file** that:

- Runs both the **publisher** and **subscriber** nodes for publishing and subscribing complex numbers.
- Allows the user to set the **complex number** parameters (real and imaginary) at launch-time.

What I Did

1. Created the **complex_number.launch.py** file inside the **lab1** package to declare parameters for **real** and **imaginary** parts of a complex number.
2. Updated the **publisher.py** to read these parameters and publish complex numbers to the **/complex_numbers** topic.
3. Launched both the publisher and subscriber nodes using the launch file.

A terminal window titled 'ahmadgill@ahmadgill: ~/ros2_ws' with a search bar and window controls. The terminal output shows the installation of the 'robotics_lab' package, followed by running 'ros2 run turtlesim turtlesim_node'. It displays information about spawning a turtle at a specific position and orientation, along with several warnings from libEGL and MESA. The background of the terminal window shows a desktop environment with a purple theme and a cat icon.

```
ahmadgill@ahmadgill: ~/ros2_ws
ros2_ws/install/robotics_lab' in the environment variable AMENT_PREFIX_PATH doesn't exist
Starting >>> robotics_lab
Finished <<< robotics_lab [0.78s]

Summary: 1 package finished [0.90s]
ahmadgill@ahmadgill:~/ros2_ws$ ros2 run turtlesim turtlesim_node
SocketNotifier: Can only be used with threads started with QThread
[INFO] [1770387503.156166515] [turtlesim]: Starting turtlesim with node name /turtlesim
[INFO] [1770387503.158839540] [turtlesim]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
libEGL warning: failed to get driver name for fd -1

libEGL warning: MESA-LOADER: failed to retrieve device information
libEGL warning: failed to get driver name for fd -1

MESA: error: ZINK: failed to choose pdev
libEGL warning: egl: failed to create dri2 screen
[INFO] [1770387918.600620215] [rclcpp]: signal_handler(SIGINT/SIGTERM)
ahmadgill@ahmadgill:~/ros2_ws$ ros2 run robotics_lab publisher
[INFO] [1770387920.794154001] [simplePublisher]: Publishing: 0.0 + 0.0j
```

Complex Number

3. Task 2: Square and Circle Movement for Two Turtles

Objective

In this task, I created two behaviors for the turtles:

1. **Turtle1** moves in a square pattern.
2. **Turtle2** moves in a circular pattern.

I also created a launch file that:

- Spawns turtle2 at a user-defined position.
- Runs both movement nodes.

What I Did

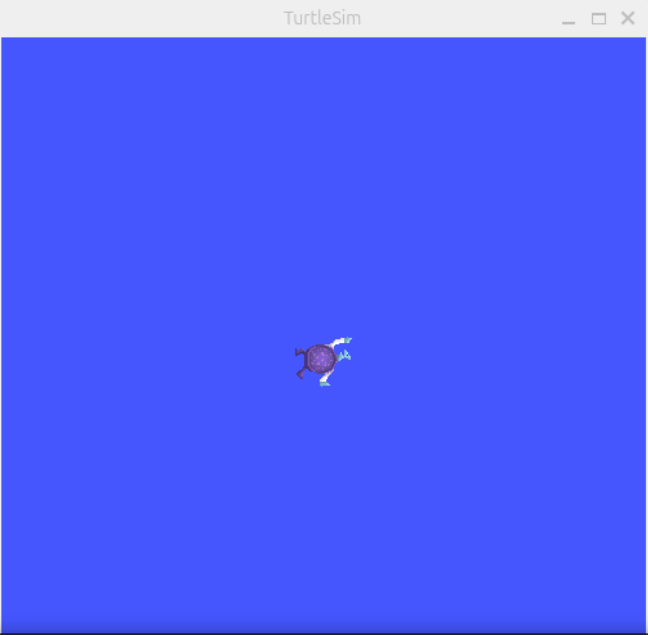
1. Created the **move_square.py** node for **turtle1** to move in a square.
2. Created the **move_circle.py** node for **turtle2** to move in a circle.
3. Created a launch file, **circle_plus_square.launch.py**, that launches both turtles and their respective movement behaviors.

```
ahmadgill@ahmadgill: ~/ros2_ws

colcon build
source /opt/ros/jazzy/setup.bash
colcon build
source install/setup.bash
Starting >>> robotics_lab
Finished <<< robotics_lab [0.78s]

Summary: 1 package finished [0.90s]
ahmadgill@ahmadgill:~/ros2_ws$ ros2 run robotics_lab moveStraight
[INFO] [1770387589.595424903] [move_straight_node]: Turtle moved straight for 2
seconds.
```

TurtleSim



```
... path '/home/ahmadgill/r
... AMENT_PREFIX_PATH doesn

sim_node
with QThread
clesim with node name /tu
le [turtle1] at x=[5.544

Information
```

Move Straight

```
ahmadgill@ahmadgill: ~/ros2_ws

TurtleSim

[INFO] [1770387589.595424903] [move_straight_node]: Turtle moved straight for 2
seconds.
```

TurtleSim



```
... Starting turtlesim
... Spawning turtle [t
... name for fd -1
... to retrieve device inform
... name for fd -1
... odev
... dri2 screen
... esim]: Spawning turtle [t
... le_node]: Turtle spawned
... le_node]: Turtle2 finish
... d 21223]
... are_node]: Turtle complet
... d 21222]
```

```
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/robotics_lab$ ls
__init__.py  moveSquare.py  moveToGoal.py
cle.py       moveStraight.py  publisher.py
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/robotics_lab$ nano chase.py
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/robotics_lab$ cd ..
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab$ cd launch/
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/launch$ nano chase_turtle.launch.
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/launch$ cd ..
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab$ cd ..
ahmadgill@ahmadgill: ~/ros2_ws/src$ ls
s_lab
ahmadgill@ahmadgill: ~/ros2_ws/src$ cd robotics_lab/
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab$ ls
package.xml  resource  robotics_lab  setup.cfg  setup.py  test
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab$ nano setup.py
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab$ ls
package.xml  resource  robotics_lab  setup.cfg  setup.py  test
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab$ cd robotics_lab/
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/robotics_lab$ ls
chase.py  moveCircle.py  moveStraight.py  publisher.py
__init__.py  moveSquare.py  moveToGoal.py
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/robotics_lab$ nano chase.py
ahmadgill@ahmadgill: ~/ros2_ws/src/robotics_lab/robotics_lab$
```

Circle and Square

Task 3: Turtle Chasing Behavior

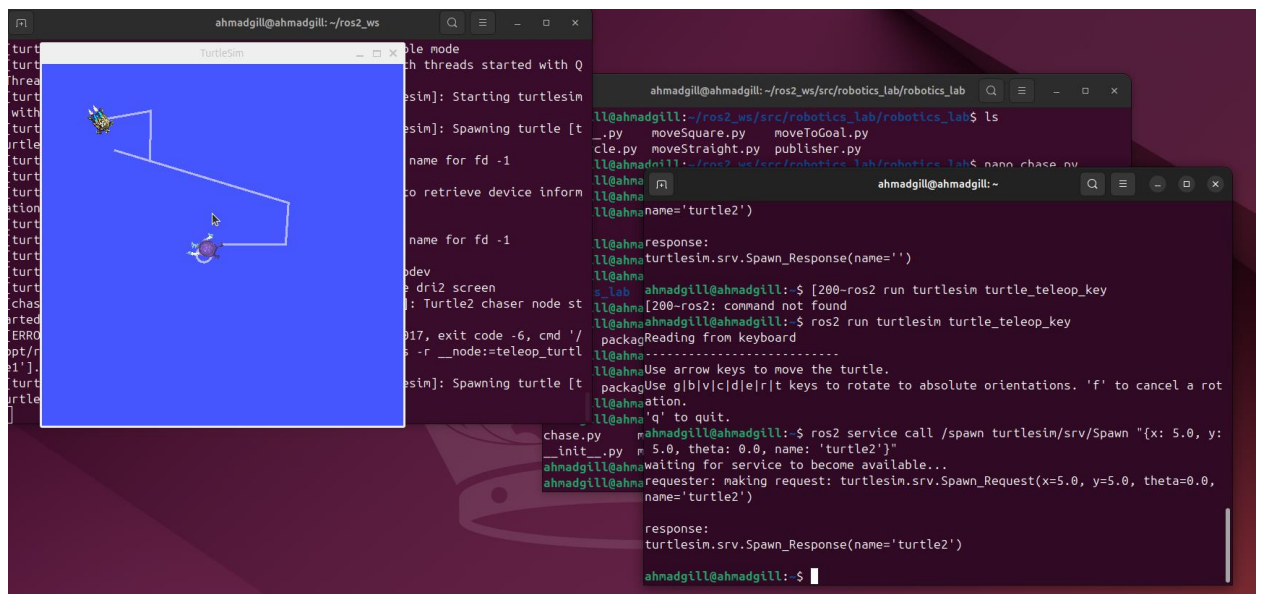
Objective

In this task, I implemented a chasing behavior where **turtle2** chases **turtle1**:

1. **Turtle 1** is controlled using keyboard teleoperation.
2. **Turtle 2** automatically chases **Turtle 1** by subscribing to its position and adjusting its velocity.

What I Did

1. Spawned **turtle2** at a specific position and controlled **turtle1** using the **keyboard** (turtle_teleop_key).
2. Created **chase.py**, where **turtle2** calculates the relative heading and distance from **turtle1** and adjusts its velocity accordingly.



Chase

